ME 165
Basic Mechanical Engineering

Lecture 10

Kinematics of Particles

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Particle Kinematics:

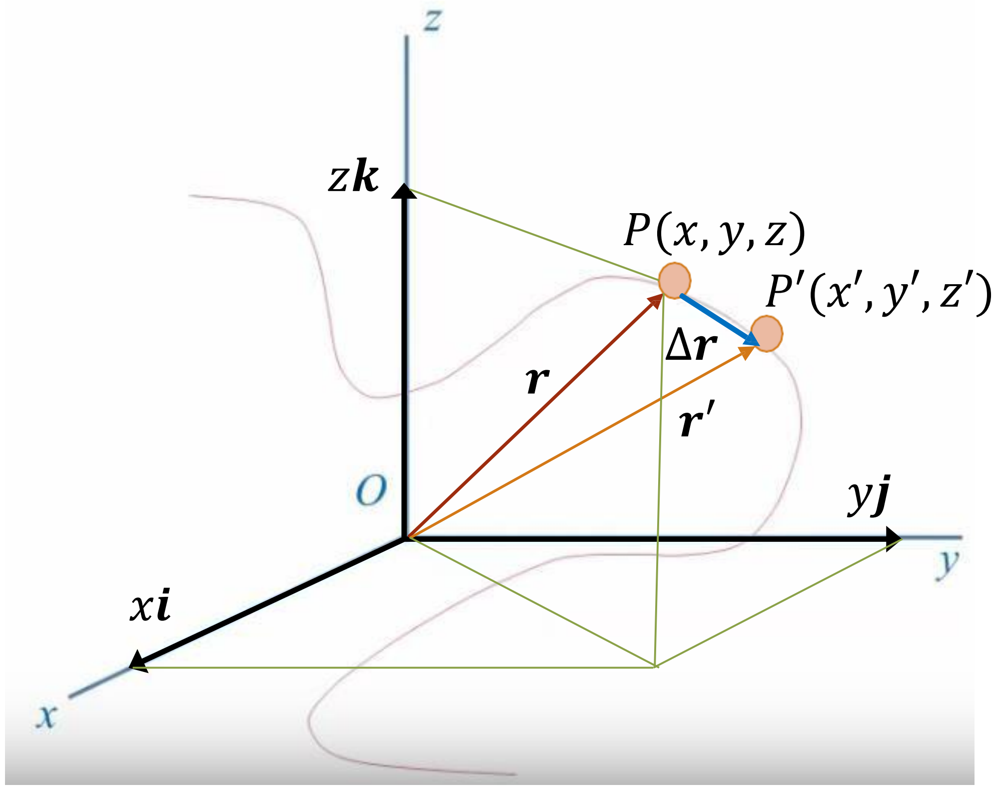
$$\boldsymbol{r} = x\boldsymbol{i} + y\boldsymbol{j} + z\boldsymbol{k}$$

$$v = \frac{dr}{dt}$$

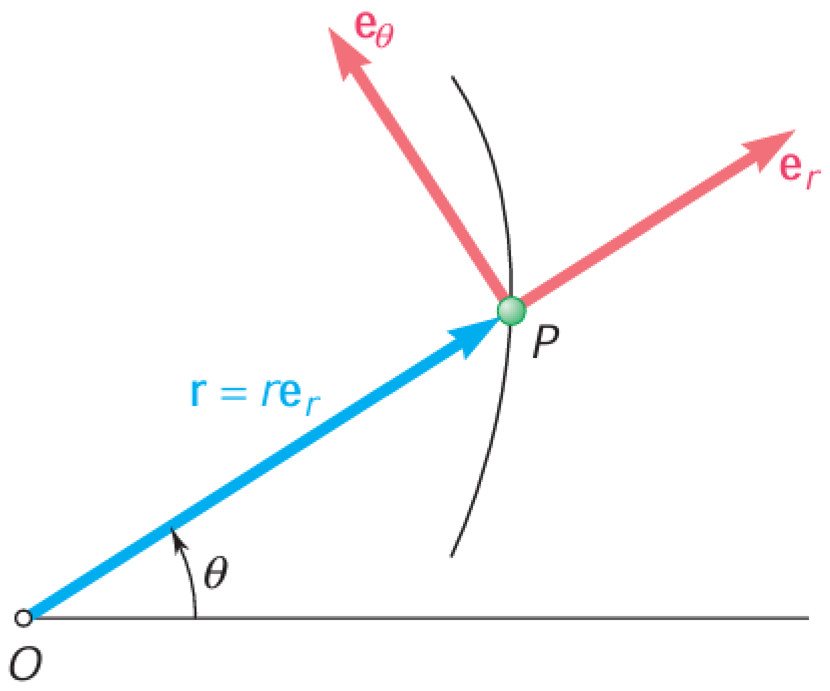
$$\therefore \boldsymbol{v} = \dot{x}\,\boldsymbol{i} + \dot{y}\,\mathbf{j} + \dot{z}\,\boldsymbol{k} = v_x\boldsymbol{i} + v_y\boldsymbol{j} + v_z\boldsymbol{k}$$

$$a = \frac{dv}{dt} = \frac{d^2r}{dt^2}$$

$$\therefore \boldsymbol{a} = \ddot{x}\,\boldsymbol{i} + \ddot{y}\,\boldsymbol{j} + \ddot{z}\,\boldsymbol{k} = a_x\boldsymbol{i} + a_y\boldsymbol{j} + a_z\boldsymbol{k}$$



Particle Kinematics: Radial & Transverse Components

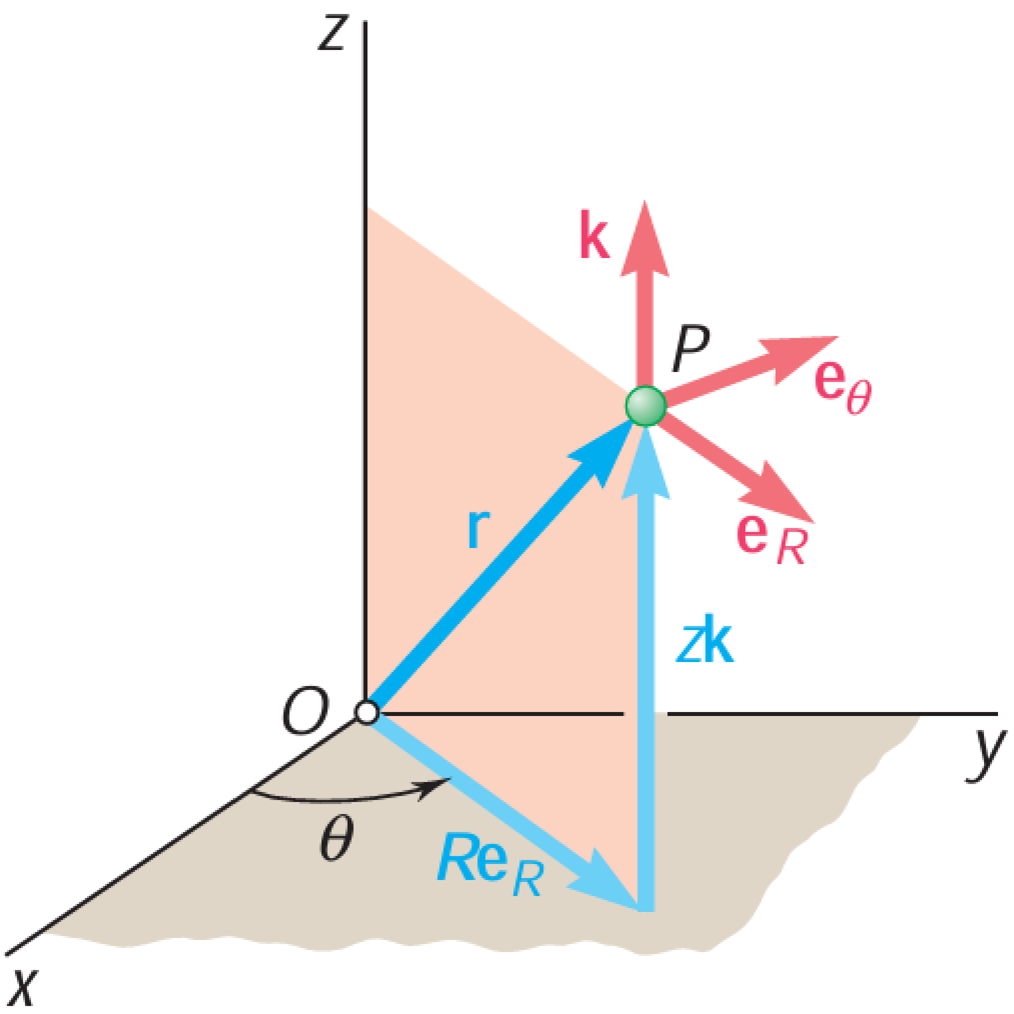


r **: Radial Coordinate**
θ **: Transverse Coordinate**

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\mathrm{u}}\mathbf{e}_\mathrm{u}$$

$$\boldsymbol{a} = (\ddot{r} - r\dot{\mathrm{u}}^2)\mathbf{e}_r + (r\ddot{\mathrm{u}} + 2\dot{r}\dot{\mathrm{u}})\mathbf{e}_\mathrm{u}$$

Particle Kinematics: Cylindrical Components



Cylindrical components

Position: $\overline{\boldsymbol{r_P}} = r\overline{\boldsymbol{u_r}} + \boldsymbol{z}\overline{\boldsymbol{u_z}}$

Velocity:

$$\overline{\boldsymbol{v}} = \frac{d\overline{\boldsymbol{r_P}}}{dt}$$

$$\overline{\boldsymbol{v}} = v_r\overline{\boldsymbol{u_r}} + v_\theta\overline{\boldsymbol{u_\theta}} + v_z\overline{\boldsymbol{u_z}}$$

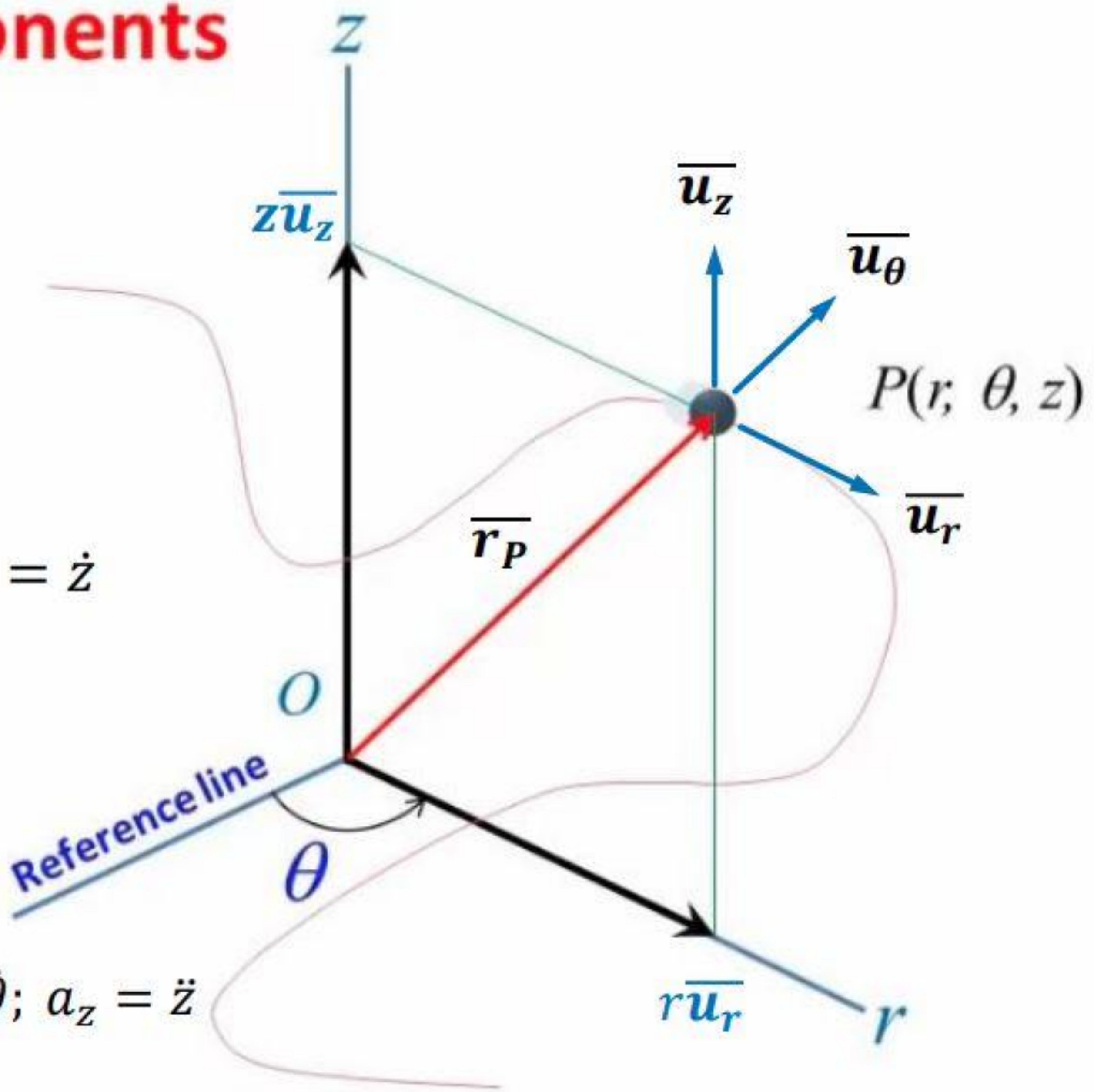
$$v_r = \dot{r}\,; \quad v_\theta = r\dot{\theta}\,; \quad v_z = \dot{z}$$

Acceleration:

$$\overline{\boldsymbol{a}} = \frac{d\overline{\boldsymbol{v}}}{dt}$$

$$\overline{\boldsymbol{a}} = a_r\overline{\boldsymbol{u_r}} + a_\theta\overline{\boldsymbol{u_\theta}} + a_z\overline{\boldsymbol{u_z}}$$

$$a_r = \ddot{r} - r\dot{\theta}^2\,;\ a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta};\ a_z = \ddot{z}$$



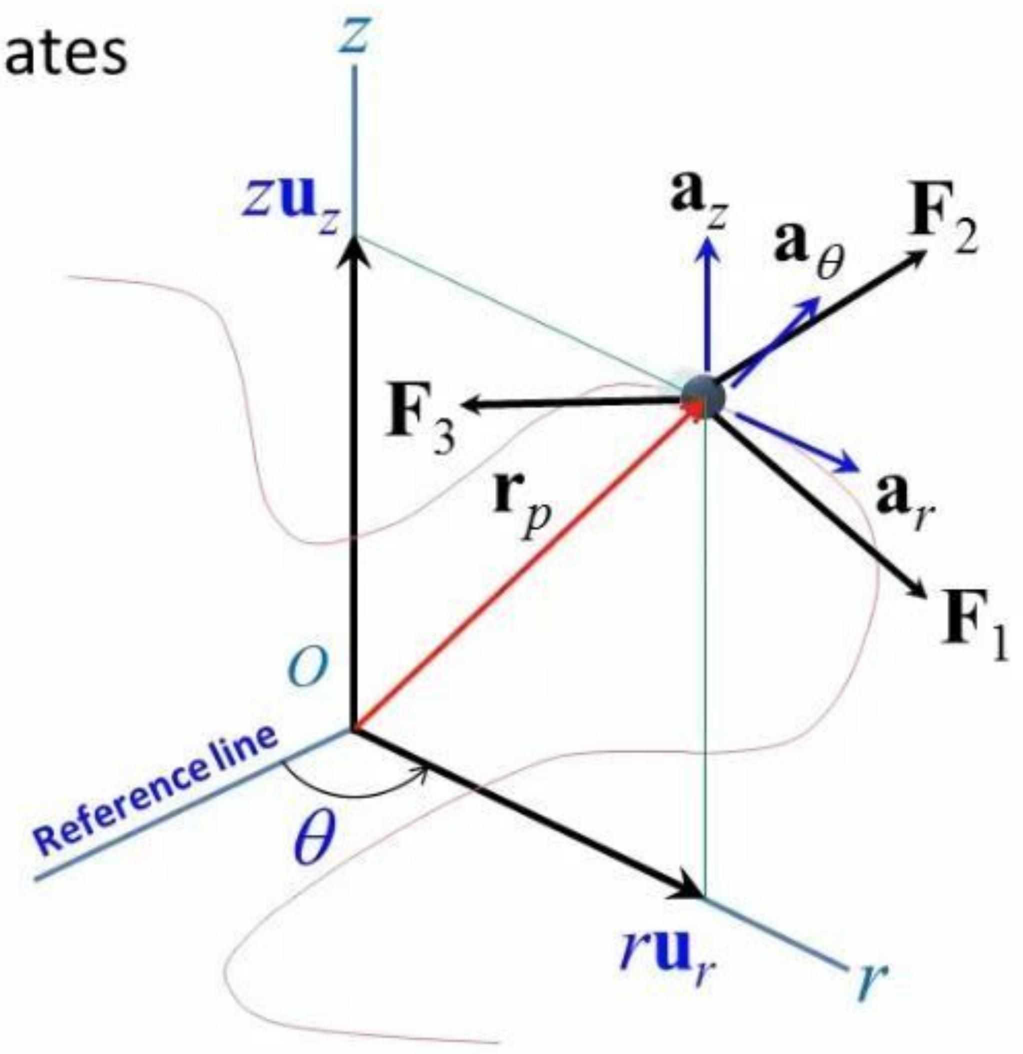
Particle Kinematics: Cylindrical Components

Cylindrical coordinates

$$\sum \mathbf{F} = m\mathbf{a}$$

$$\begin{cases} \sum F_r = ma_r \\ \sum F_\theta = ma_\theta \\ \sum F_z = ma_z \end{cases}$$

3D



Particle Kinematics: Cylindrical Components

Polar coordinates

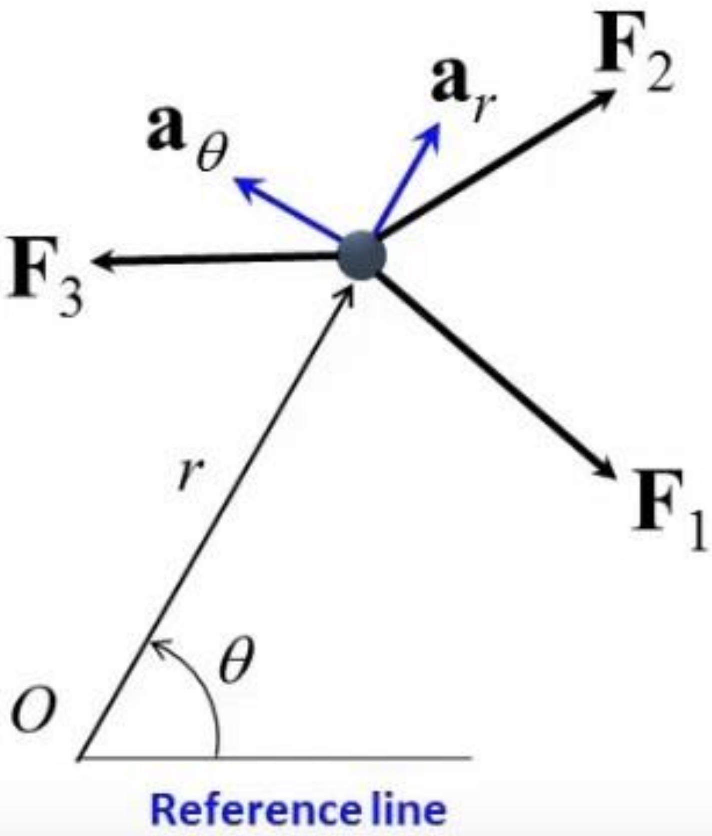
$$\sum \mathbf{F} = m\mathbf{a}$$

$$\boxed{\begin{cases} \sum F_r = ma_r \\ \sum F_\theta = ma_\theta \end{cases}}$$

2D

$$a_r = \ddot{r} - r\dot{\theta}^2$$

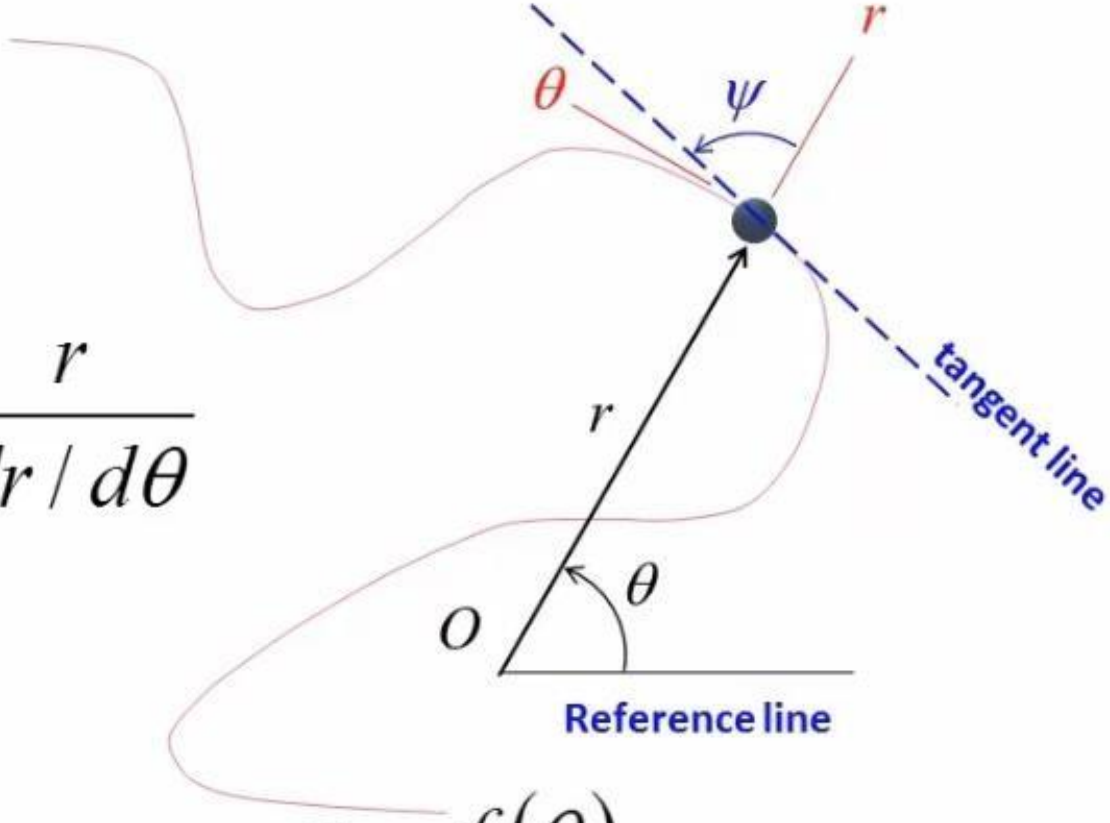
$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$



Particle Kinematics: Cylindrical Components

Cylindrical coordinates

$$\psi = \tan^{-1} \frac{r}{dr / d\theta}$$



$$r = f(\theta)$$

Particle Kinematics: Cylindrical Components

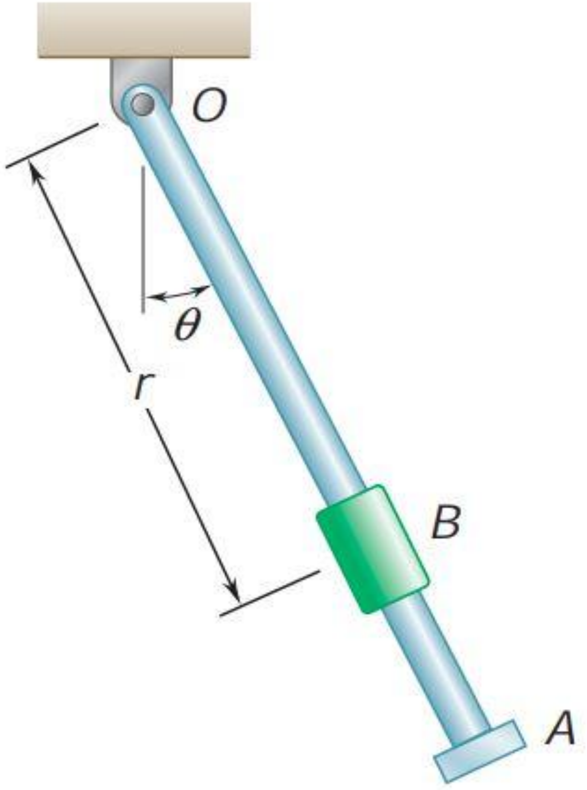
Problem:

The oscillation of rod OA about O is defined by the relation $\boldsymbol{\theta} = \left(\frac{\mathbf{3}}{\boldsymbol{\pi}}\right) \mathbf{sin}(\boldsymbol{\pi t})$, where θ and t are expressed in radians and seconds, respectively. Collar B slides along the rod so that its distance from O is **r = 6(1-$\boldsymbol{e^{-2t}}$)** where r and t are expressed in inches and seconds, respectively. When t =1s, determine

(a) the velocity of the collar,

(b) the acceleration of the collar,

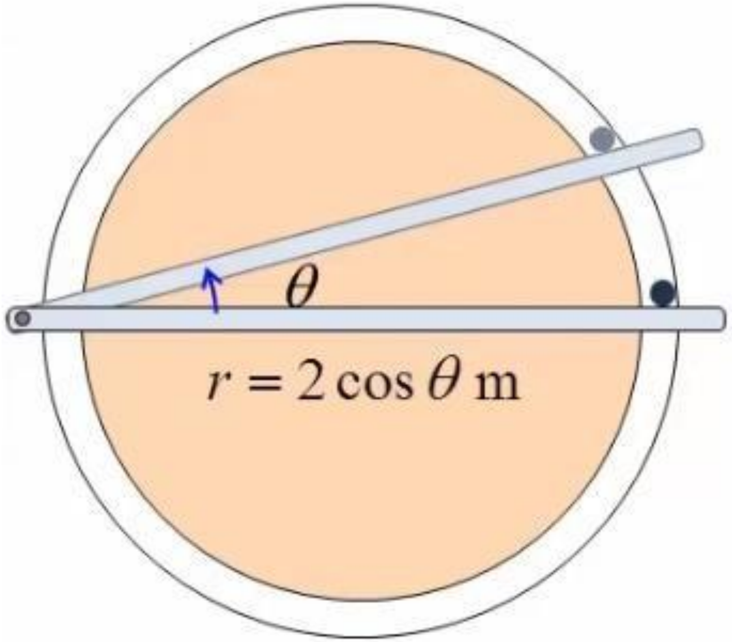
(c) the acceleration of the collar relative to the rod.



Particle Kinematics: Cylindrical Components

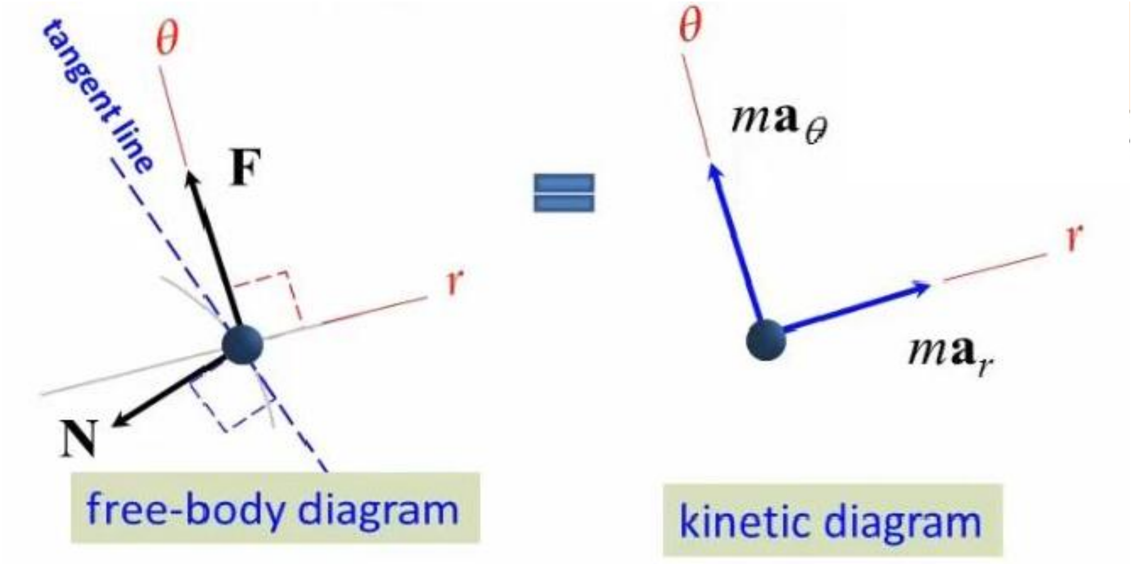
Problem:

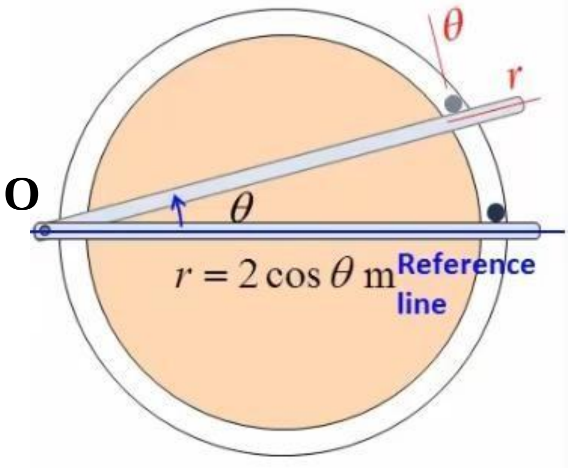
A 1-kg ball is being pushed by a rod to move in a horizontal grooved, smooth slot. It starts from $\theta = 0°$. Determine the force the rod exerts on the ball at $\theta = 15°$, if at this instant the rod moves at angular $\dot{\theta} = 1\, rad/s$, with angular acceleration $\ddot{\theta} = 2\, rad/s^2$. The ball is only in contact with the outer side of the slot.



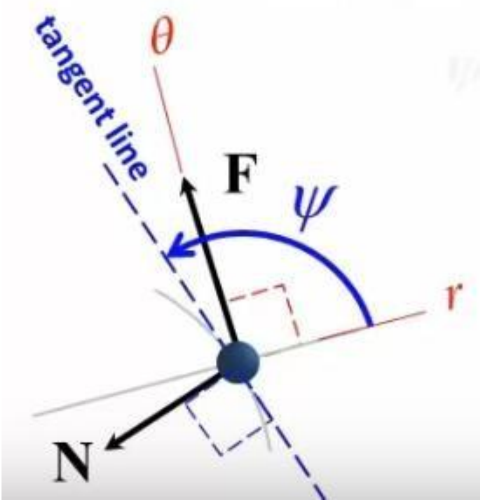
Equation of Motion : Cylindrical Components

Solution:





$$\frac{dr}{d\theta} = -2\sin\theta$$



$$\psi = \tan^{-1}\frac{r}{dr/d\theta}$$

at $\theta = 15°$:

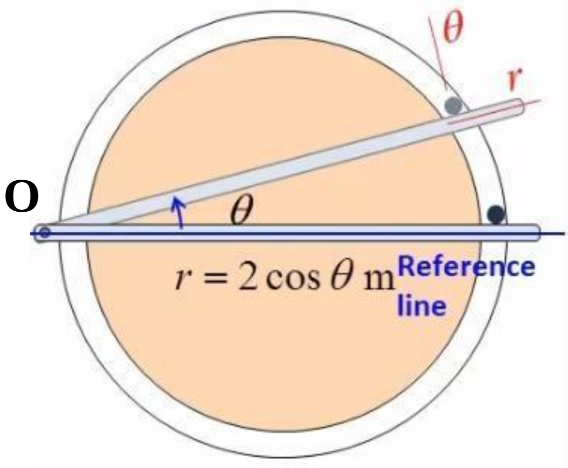
$$\psi = \tan^{-1}\frac{2\cos 15°}{-2\sin 15°} = -75°$$

$$\sum F_r = -N\sin 75° = ma_r$$

$$\sum F_\theta = F - N\cos 75° = ma_\theta$$

Equation of Motion : Cylindrical Components

Solution:



$$r = 2\cos\theta$$

$$\dot{r} = \frac{dr}{dt} = \frac{dr}{d\theta}\cdot\frac{d\theta}{dt} = -2\sin\theta\dot{\theta}$$

$$\ddot{r} = \frac{d\dot{r}}{dt} = -2\sin\theta\frac{d\dot{\theta}}{dt} - 2\frac{d\sin\theta}{dt}\dot{\theta} = -2\sin\theta\ddot{\theta} - 2\frac{d\sin\theta}{d\theta}\cdot\frac{d\theta}{dt}\cdot\dot{\theta}$$

$$= -2\sin\theta\ddot{\theta} - 2\cos\theta\cdot\dot{\theta}^2$$

at $\theta = 15°$: $\quad \dot{\theta} = 1 \quad \ddot{\theta} = 2$

$$r = 2\cos\theta = 1.932$$

$$\dot{r} = -2\sin\theta\dot{\theta} = -0.5176$$

$$\ddot{r} = -2\sin\theta\ddot{\theta} - 2\cos\theta\cdot\dot{\theta}^2 = -2.967$$

$$a_r = \ddot{r} - r\dot{\theta}^2 = -4.899$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 2.828$$

$\overset{+}{\nearrow}$ $\sum F_r = -N\sin 75° = ma_r$

$\overset{+}{\nwarrow}$ $\sum F_\theta = F - N\cos 75° = ma_\theta$

$$\therefore \begin{cases} N = 5.07\ \text{N} \\ F = 4.14\ \text{N} \end{cases}$$